Critical Success Factors for Gold-Backed Stablecoins Development in Hong Kong*

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Abstract

Against the structural transformation of the global monetary system, weakened sovereign credit, accelerated gold re-monetization, and the rise of real-world asset tokenization, gold-backed stablecoins have become an important innovation in digital finance and cross-border payments. This study employs a systematic literature review to identify and analyze the critical success factors for the development of gold-backed stablecoins in Hong Kong from seven dimensions: collateral quality, regulatory certainty, reserve governance, price stability mechanisms, locational-ecological synergy, risk prevention systems, and compliance-technology integration. The findings indicate that Hong Kong's triple synergies as a gold hub, offshore financial center, and digital asset center, together with full-cycle risk governance and technology-embedded compliance, form the core logic for the sustainable development of gold-backed stablecoins. This study contributes to theoretical understanding of commodity-pegged stablecoins and provides practical implications for regulatory design and institutional development in offshore financial centers.

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1. Introduction

The global monetary system is undergoing its most profound structural transformation since the end of the Cold War. Volatility in U.S. monetary policy, expanding federal debt, infringements on monetary policy independence, and the instrumentalization of the U.S. dollar in cross-border payments and reserve systems have driven central banks and institutional investors worldwide to restructure asset allocations. Gold has re-emerged as a core supra-sovereign reserve asset, entering a new phase of re-monetization [1][2]. In 2024, global central banks were net purchasers of gold for the third consecutive year, exceeding 1,000 tonnes, raising gold's share in international reserves to 20%, surpassing the euro to become the second-largest reserve asset after the U.S. dollar [1][2].

Meanwhile, stablecoins have rapidly evolved from auxiliary tools for cryptocurrency trading into instruments for cross-border payments, commodity settlement, asset management, supply chain finance, and trade financing, with annual transaction volumes exceeding one trillion U.S. dollars, becoming critical infrastructure for the global internet of value [3][9][22]. During rapid expansion, the stablecoin industry has repeatedly experienced systemic risks, including death spirals in algorithmic stablecoins, reserve shortages and transparency crises in some fiat-collateralized varieties, and bank runs triggered by failed redemption mechanisms [3][12][21].

These events repeatedly demonstrate that the long-term success of a stablecoin depends on the reliability of collateral assets, reserve transparency, regulatory certainty, and risk governance capacity [3][4][21]. Compared with fiat-collateralized, crypto-collateralized, and algorithmic stablecoins, Gold-Backed Stablecoins (GBSCs) use standardized, highly liquid, and globally recognized physical gold as underlying collateral. They do not rely on single sovereign credit and carry inherent safe-haven attributes, making them more suitable for cross-border trade, wealth management, foreign exchange reserve diversification, and hedging geopolitical risk, thus representing a significant direction for next-generation compliant stablecoins [2][19][23].

Hong Kong provides one of the world's most supportive environments for Gold-Backed Stablecoin development. As an international financial center, Asia's hub for gold spot pricing and trading, and the world's largest offshore Renminbi center, Hong Kong possesses a complete ecosystem for gold storage, clearing, settlement, and auditing. The Stablecoin Ordinance, formally implemented in 2024, establishes a comprehensive regulatory framework including licensing, reserve segregation, independent auditing, swift redemption, and investor protection, making Hong Kong one of the few jurisdictions globally to provide clear legal status for commodity-backed stablecoins [8][15][21]. Nevertheless, Gold-Backed Stablecoins still face multiple constraints, including gold price volatility that may undermine peg stability, large-scale redemptions that may trigger liquidity pressures, reserve authenticity and custodial security that shape market trust, cross-border compliance and anti-money laundering requirements that raise operational thresholds, and technological security and on-chain verification efficiency that limit scalability [3][12][21]. Therefore, systematically identifying the critical success factors for Gold-Backed Stablecoin development in Hong Kong is both an important theoretical proposition in digital finance, monetary economics, and regulatory technology, and a practical necessity for Hong Kong's goal of building an international digital asset center and consolidating its status as a global gold hub.

2. Literature Review

2.1 Current State of Stablecoins

Existing literature predominantly focuses on fiat-backed stablecoins, particularly USDT, which accounts for approximately 62% of global stablecoin market capitalization and processes over $120 billion in daily trading volume [8][15]. Empirical studies indicate that the success of fiat-backed stablecoins rests on early-mover advantages, reserve liquidity, multi-chain deployment, and arbitrage efficiency [10][12][22]. However, almost no empirical research targets gold-backed stablecoins, especially

within an offshore financial center such as Hong Kong. Three landmark academic studies confirm this critical gap: first, research shows that over 90% of stablecoin scholarship focuses on USD-pegged models, with no structured framework for commodity-backed designs [4][8]; second, redemption accessibility, reserve transparency, and regulatory clarity explain 78% of adoption variance, yet gold-backed structures remain largely untested [10][22]; third, asset volatility, custody risk, cross-border regulatory inconsistency, and institutional acceptance represent the top barriers to scaling non-fiat stablecoins [19][24].

2.2 Hong Kong faces distinct challenges for GBSCs

Gold-backed stablecoins in Hong Kong further face distinct challenges, including gold price fluctuations that disrupt stability mechanisms, strict compliance requirements under the Stablecoin Ordinance that increase operational costs, custody and physical verification expenses that hinder scalability, cross-border capital supervision that restricts instant circulation, and low institutional awareness relative to USD stablecoins [21][15][20]. This study addresses two core research questions: what are the critical factors for sustainable success of Gold-Backed Stablecoins within Hong Kong's institutional and market environment, and how do these factors interact to form a stable and self-reinforcing support system. The academic contributions of this paper are threefold: first, it constructs a seven-dimensional framework of success factors for Gold-Backed Stablecoins, addressing a major gap in research on commodity-pegged varieties; second, it analyzes synergies between the Stablecoin Ordinance and Hong Kong's gold ecosystem, revealing the institutional logic for offshore financial centers to support compliant stablecoin innovation [15]; third, it establishes an integrated analytical paradigm of asset–regulation–mechanism–ecosystem–risk control–technology, providing a reusable framework for global research on commodity-pegged stablecoins.

2.3 Stablecoin Development Status

Stablecoins have evolved into a foundational pillar of modern digital finance. By 2024, the total market capitalization of stablecoins exceeded $130 billion, with USD-pegged instruments representing more than 90% of the entire market [4][8]. USDT alone accounts for 62% of decentralized trading volume and supports over $120 billion in daily transactions, underscoring its role as critical market infrastructure [10][15]. However, commodity-backed stablecoins represent less than 2% of total market capitalization, indicating massive, untapped potential and a clear need for deeper research into their success conditions [19][24]. Gold-backed stablecoins represent a distinct and innovative subclass that digitally links on-chain tokens to physical gold reserves, and gold's 20% share in global reserves strengthens its credibility as collateral [1][2]. Hong Kong's position as a global gold trading hub, combined with its comprehensive Stablecoin Ordinance, creates a unique living laboratory for Gold-Backed Stablecoin development, testing, and scaling [15][20][21].

2.4 Gold-Backed Stablecoins and Hong Kong's Context

The essence of a stablecoin is a value-pegging instrument supported by collateral assets and formal governance mechanisms, and scholars categorize stablecoins into fiat-collateralized, commodity-collateralized, and algorithmic types, with consistent findings that asset-collateralized stablecoins exhibit significantly greater stability [21][9]. Stability depends on adequate reserves, credible redemption, and transparent governance, while major failures stem from insufficient reserves, asset misappropriation, information opacity, flawed redemption mechanisms, and weak price stabilization[12][21]. Gold-Backed Stablecoins fall into the commodity-collateralized category, and their stability directly depends on gold asset quality, collateral ratios, reserve management, and arbitrage efficiency[23][22]. Gold possesses four defining characteristics—supra-sovereignty, zero default risk, inflation resistance, and low cross-asset correlation—that make it a natural monetary anchor [21][1]. During periods of rising sovereign currency uncertainty, gold resumes its role as a reserve and payment anchor in a process known as re-monetization[1][2]. Gold-Backed Stablecoins represent a digitized, tokenized, and on-chain representation of gold, preserving its value attributes while extending payment

and settlement functionality to enable fractional ownership, programmable transactions, and real-time cross-border transfers [24][22].

The core competitiveness of an offshore financial center stems from regulatory certainty, market openness, asset liquidity, legal reliability, and cross-border connectivity[20][5]. The literature widely confirms that Hong Kong possesses comprehensive advantages in stablecoin regulation, gold market infrastructure, cross-border payment systems, and institutional investor depth, which collectively provide the foundations for trust, liquidity, and scalability in Gold-Backed Stablecoins [20][21][15]. Effective regulation is a necessary prerequisite for scalable and sustainable stablecoin development, and a robust framework must prevent systemic risk, protect token holders, and promote responsible compliant innovation [29][5]. Hong Kong's Stablecoin Ordinance achieves these goals through licensing requirements, capital adequacy rules, reserve standards, disclosure obligations, and risk control mechanisms [15][21]. This framework provides legal certainty for Gold-Backed Stablecoins, reduces policy uncertainty, and enhances institutional willingness to participate, striking a balanced approach that supports innovation while safeguarding financial stability [8][21].

Existing literature extensively discusses stablecoin risks, regulatory design, and mechanism engineering, yet systematic research on success factors for Gold-Backed Stablecoins in offshore financial centers remains limited, particularly lacking integrated analysis of asset pegging, regulation, reserve governance, price stability, locational ecology, risk control, and technology-compliance integration [21][20][22]. Most studies focus on fiat or algorithmic models rather than commodity-backed alternatives, and few account for Hong Kong's unique triple advantages as a gold hub, offshore financial center, and digital asset center.

Hong Kong's Stablecoin Ordinance provides clear legal status, operational rules, and trust foundations for Gold-Backed Stablecoins, with critical provisions including a licensing system with stringent entry thresholds, mandatory 100% reserve coverage of circulating stablecoin value, legal segregation and third-party custody of reserves to protect against issuer bankruptcy, independent auditing and daily disclosure requirements, and explicit compliance pathways for commodity-pegged instruments [15][21][28]. This framework aligns with the EU's MiCAR regulation while adapting to Asian market structures and gold asset characteristics, serving as the institutional cornerstone for safe and sustainable Gold-Backed Stablecoin development [8][21].

Hong Kong possesses one of the world's most complete and mature gold industry ecosystems, serving as Asia's leading center for gold spot pricing, trading, and clearing with deep liquidity and globally recognized benchmarks [20][1]. It supports a mature system of licensed vaults, secure logistics, professional assay services, and independent auditing firms that ensure gold authenticity and integrity [2][20]. The city benefits from unrestricted capital flow, no foreign exchange controls, and a common-law legal system aligned with international standards, which lowers frictions for cross-border gold and stablecoin transactions [20][21]. It hosts a dense cluster of institutional investors with strong and growing demand for cross-border payments, wealth management, and alternative safe-haven assets [20][5]. Hong Kong's physical gold storage network is among the most secure and cost-effective in Asia, allowing issuers to minimize custody costs while maximizing investor confidence, and its continuous 24-hour gold pricing ecosystem linking London, New York, and Shanghai provides reliable and timely price feeds for on-chain oracle integration.

Hong Kong has established a synergistic landscape that includes virtual asset licensing, regulated exchanges, institutional custody, and enterprise-grade blockchain infrastructure, providing strong technical and market support for gold tokenization and stablecoin operations[24][39]. This ecosystem bridges traditional financial institutions and digital asset platforms, enabling institutional investors to access Gold-Backed Stablecoins through familiar, regulated channels, while also supporting complementary services such as on-chain proof-of-reserves, automated auditing, and institutional-grade wallet solutions that strengthen transparency and trust.

3. Methodology

This study adopts a systematic literature review method, which is widely recognized in top finance and digital asset journals for synthesizing theoretical and empirical evidence to identify critical success factors. The systematic literature review approach ensures rigor, transparency, and replicability by following structured procedures for literature identification, screening, evaluation, and synthesis, allowing integration of qualitative theories, quantitative empirical findings, and regulatory insights into a unified framework suitable for exploring emerging topics such as gold-backed stablecoins in a novel institutional setting. Literature collection covers high-quality academic papers, industry reports, regulatory documents, and case studies from authoritative databases including Google Scholar, Web of Science, Scopus, CNKI, SSRN, and official publications from the Hong Kong Monetary Authority, with search terms including gold-backed stablecoins, stablecoin success factors, Hong Kong stablecoin regulation, commodity-pegged stablecoins, gold re-monetization, and digital asset hubs.

Data analysis proceeds in three stages: descriptive and thematic analysis to classify literature by research focus, theoretical perspective, and empirical context; critical content analysis to extract core variables, causal mechanisms, success conditions, and risk factors; and framework synthesis to integrate extracted factors into a cohesive seven-dimensional model with logical linkages between institutional conditions, market structures, technological mechanisms, and governance arrangements, ensuring both interpretive depth and structural rigor consistent with top-tier academic publication standards.

4. Findings

The primary prerequisite and foundational value driver for Gold-Backed Stablecoin success is that the collateral asset itself carries global consensus, low risk, high liquidity, and verifiability. Gold carries no sovereign default risk and does not depend on any government, central bank, or corporate credit, maintaining resilient value during geopolitical conflict, sanctions, or currency devaluation—an advantage unmatched by fiat currencies[21]. Gold's long-term volatility is significantly lower than that of crypto-assets, and it exhibits a low correlation with equity and bond markets, thereby reducing underlying volatility risks for stablecoin holders [23]. Gold benefits from a globally unified, continuous, and transparent 24-hour pricing mechanism across London, Shanghai, and New York, providing reliable inputs for on-chain oracles [22]. It is highly standardized, verifiable, and custody-friendly, making it well-suited for on-chain representation and proof-of-reserves[24]. Hong Kong's mature gold infrastructure further ensures physical existence, traceability, and deliverability, eliminating trust crises related to fictitious reserves at the source [20].

The institutional prerequisite for scalable and institutional Gold-Backed Stablecoin development in Hong Kong is a clear, stable, predictable, and internationally aligned regulatory framework, which provides clear legality including commodity-pegged stablecoins within regulated boundaries, a licensing system that reduces adverse selection by raising entry thresholds, mandatory 100% reserves that eliminate fractional-reserve risks, legal segregation and custody that protect holder assets, and statutory redemption rights that guarantee the ultimate price-stabilizing mechanism [15][21][28]. Regulatory certainty delivers a triple effect: institutional capital is more willing to enter, cross-border cooperation proceeds smoothly, and market trust forms rapidly, with the literature indicating that regulatory transparency and rule stability are critical determinants of successful financial innovation [8][5].

Trust lies at the core of stablecoin functionality, and it arises from verifiable constraint mechanisms, requiring Gold-Backed Stablecoins to build a trust triangle comprising full collateralization, reserve segregation, and independent auditing. They require 100% or greater collateralization with physical gold, prohibiting fractional reserves, leveraged exposure, or substitution with risky assets [21]. Reserve assets must be legally segregated and held by licensed Hong Kong trustees or banks, achieving robust bankruptcy remoteness [28]. A dual verification system combines on-chain Proof-of-Reserves, off-site

physical verification, and third-party independent auditing, while comprehensive disclosure of gold storage location, weight, purity, and transaction records enables penetrative supervision and public accountability [22][21]. The literature indicates that insufficient, opaque, or misappropriated reserves are the leading causes of stablecoin failures [12][21]. The core function of a Gold-Backed Stablecoin is to maintain a reliable peg, which depends on three synergistic on-chain and off-chain mechanisms: decentralized oracles provide manipulation-resistant, real-time, multi-source aggregated gold prices; smart contracts maintain dynamic over-collateralization ratios of at least 120% to absorb short-term fluctuations; an open minting-redemption mechanism enables risk-free arbitrage to restore price parity; and circuit breakers, redemption limits, and liquidity buffers prevent contagious bank runs under extreme conditions [22][21][12]. Literature proves that collateralized stablecoin stability depends on effective arbitrage, and gold's high liquidity makes this adjustment fast and reliable [22][13].

Scalability and sustainability of Gold-Backed Stablecoins flow from ecosystem-level network effects, and Hong Kong possesses a globally unique combination of advantages: a complete gold ecosystem covering pricing, storage, clearing, settlement, assay, and auditing; an offshore financial ecosystem featuring free capital movement, efficient cross-border payments, and dense institutional clustering; a digital asset ecosystem supported by clear regulation, comprehensive licensing, and mature infrastructure; and a strategic connection between Mainland China and global markets backed by the world's largest gold production and consumption base [20][21][15].

This multi-layered synergy allows Gold-Backed Stablecoins to simultaneously achieve physical backing, on-chain efficiency, cross-border liquidity, and institutional-grade trust. Gold-Backed Stablecoins must address five major risks—gold price volatility, bank runs, liquidity dry-ups, compliance breaches, and cyberattacks—requiring a full-cycle risk governance system that includes run prevention through full collateralization, swift redemption, and liquidity pools; price volatility management via over-collateralization, dynamic adjustment, and stress testing; compliance protection by embedding KYC/AML and sanctions screening into smart contracts; cybersecurity through multi-signature schemes, cold storage, and quantum-resistant encryption; and operational safety via vault security, physical verification, and internal checks and balances [12][23][29][24][20]. Literature indicates that risk governance capacity determines whether a stablecoin can survive full market cycles [21].

Long-term competitiveness of Gold-Backed Stablecoins stems from rule codification, compliance automation, and governance transparency, with requirements under the Stablecoin Ordinance implementable via smart contracts for automated reserve checks, redemption execution, on-chain disclosure, and immutable auditability[39]. This regulation–technology–market synergy meets regulatory requirements while improving efficiency and reducing trust costs, enabling continuous compliance, real-time transparency, and scalable governance for sustainable institutional adoption [39][8]. The seven success factors do not operate independently but form a mutually reinforcing, dynamically adaptive, and indispensable systemic steady-state. The gold asset provides foundational value and credibility, regulation provides institutional certainty and legal protection, reserve governance delivers trust and accountability, the price mechanism maintains stability and usability, the ecosystem provides scale, liquidity, and network effects, risk control ensures resilience and survival through stress, and technology-compliance fusion delivers efficiency, transparency, and scalability. Each factor strengthens the others, creating a robust, self-sustaining system that supports long-term success of Gold-Backed Stablecoins in Hong Kong.

5. Discussions

This study reveals that the sustainable success of Gold-Backed Stablecoins in Hong Kong depends on a holistic support system composed of seven core factors: the natural quality advantage of gold as a supra-sovereign collateral asset, regulatory certainty and compliance frameworks provided by Hong Kong's Stablecoin Ordinance, the trust triangle of full physical collateral, reserve segregation, and independent auditing, the triple price stability loop of oracle, arbitrage, and redemption, Hong Kong's

locational-ecological synergy as a gold hub, offshore financial center, and digital asset center, a full-cycle multi-layered risk governance system, and a governance model deeply integrating compliance and technology [21][15][22][20][39]. These seven factors interact dynamically and reinforce one another, collectively determining whether a Gold-Backed Stablecoin can achieve price stability, reserve credibility, institutional adoption, cross-border expansion, and long-term viability.

5.1 Supra-Sovereign Collateral Quality: The Natural Advantage of Gold

The primary prerequisite and foundational value driver for GBSC success is that the collateral asset itself carries global consensus, low risk, high liquidity, and verifiability. Gold carries no sovereign default risk and does not depend on any government, central bank, or corporate credit, maintaining resilient value during geopolitical conflict, sanctions, or currency devaluation—an advantage unmatched by fiat currencies[21]. Gold's long-term volatility is significantly lower than that of crypto-assets, and it exhibits a low correlation with equity and bond markets, thereby reducing underlying volatility risks for stablecoin holders [23]. Gold benefits from a globally unified, continuous, and transparent 24-hour pricing mechanism across London, Shanghai, and New York, providing reliable inputs for on-chain oracles[22]. It is highly standardized, verifiable, and custody-friendly, making it well-suited for on-chain representation and proof-of-reserves[24]. Hong Kong's mature gold infrastructure further ensures physical existence, traceability, and deliverability, eliminating trust crises related to fictitious reserves at the source[20]. Literature consistently confirms that pegged asset quality determines long-term trust and viability [21].

5.2 Regulatory Certainty Under Hong Kong's Stablecoin Ordinance

The institutional prerequisite for scalable and institutional GBSC development in Hong Kong is a clear, stable, predictable, and internationally aligned regulatory framework. The framework provides five core benefits: clear legality that includes commodity-pegged stablecoins within regulated boundaries[15], a licensing system that reduces adverse selection by raising entry thresholds[15], mandatory 100% reserves that eliminate fractional-reserve risks[21], legal segregation and custody that protect holder assets[28], and statutory redemption rights that guarantee the ultimate price-stabilizing mechanism[21]. Regulatory certainty delivers a triple effect: institutional capital dares to enter, cross-border cooperation proceeds smoothly, and market trust forms rapidly[21]. Academic literature indicates that regulatory transparency and rule stability are critical determinants of successful financial innovation, and Hong Kong provides a global best-practice example [8][5].

5.3 Trust Triangle: Full Collateral, Reserve Segregation, and Independent Auditing

Trust lies at the core of stablecoin functionality, and trust arises from verifiable constraint mechanisms. GBSCs must build a trust triangle composed of full collateralization, reserve segregation, and independent auditing. They require 100% or over-collateralization by physical gold, prohibiting fractional reserves, leveraged exposure, or substitution by risky assets [21]. Reserve assets must be legally segregated and held by licensed Hong Kong trustees or banks, achieving robust bankruptcy remoteness[28]. A dual verification system combines on-chain Proof-of-Reserves (PoR), off-site physical verification, and third-party independent auditing [22]. Comprehensive disclosure of gold storage locations, weights, purities, and transaction records enables penetrative supervision and public accountability [21]. The literature shows that insufficient, opaque, or misappropriated reserves are the leading causes of stablecoin failures [12][21].

5.4 Price Stability: Oracle + Arbitrage + Redemption Loop

The core function of GBSC is to maintain a reliable peg, which depends on three synergistic on-chain and off-chain mechanisms. Decentralized oracles provide manipulation-resistant, real-time, multi-source aggregated gold prices, avoiding single-point manipulation and data anomalies [22]. Smart

contracts maintain dynamic over-collateralization ratios, typically at least 120%, to absorb short-term gold price fluctuations [21]. An open minting-redemption mechanism enables risk-free arbitrage: when the stablecoin trades above parity, the market mints new tokens; when it trades below, holders redeem for the underlying assets, pushing prices back to target [22]. Under extreme market conditions, circuit breakers, redemption limits, and liquidity buffers prevent contagious bank runs[12]. Literature proves that collateralized stablecoin stability depends on effective arbitrage, and gold's high liquidity makes this adjustment fast and reliable [22][13].

5.5 Locational-Ecological Synergy: Gold Hub + Offshore Financial Center + Digital Asset Center

Scalability and sustainability of GBSCs flow from ecosystem-level network effects. Hong Kong possesses a globally unique combination of advantages: a complete gold ecosystem covering pricing, storage, clearing, settlement, assay, and auditing [20], an offshore financial ecosystem featuring free capital movement, efficient cross-border payments, and dense institutional clustering [21], a digital asset ecosystem supported by clear regulation, comprehensive licensing, and mature infrastructure[15], and a strategic connection between Mainland China and global markets, backed by the world's largest gold production and consumption base while remaining open to international investors[20]. This multi-layered synergy allows GBSCs to simultaneously achieve physical backing, on-chain efficiency, cross-border liquidity, and institutional-grade trust.

5.6 Full-Cycle Multi-Layer Risk Governance System

GBSCs must address five major risks: gold price volatility, bank runs, liquidity dry-ups, compliance breaches, and cyberattacks. A full-cycle risk governance system is therefore essential. Run prevention relies on full collateralization, swift redemption, and dedicated liquidity reserve pools[12]. Price volatility management uses over-collateralization, dynamic collateral adjustment, and regular stress testing [23]. Compliance protection embeds KYC/AML and sanctions screening directly into smart contracts[29]. Cybersecurity depends on multi-signature schemes, cold storage, on-chain monitoring, and quantum-resistant encryption[24]. Operational safety is ensured through vault security, physical verification, and internal checks and balances[20]. Literature indicates that risk governance capacity determines whether a stablecoin can survive full market cycles [21].

5.7 Compliance-Technology Integrated Governance

Long-term competitiveness of GBSCs stems from rule codification, compliance automation, and governance transparency. Requirements under Hong Kong's Stablecoin Ordinance can be implemented through smart contracts: automated reserve adequacy checks, automatic redemption execution, on-chain disclosure, and immutable data auditability [39]. This regulation–technology–market synergy meets regulatory requirements while improving efficiency and reducing trust costs [39][8]. By embedding legal obligations into code, GBSCs achieve continuous compliance, real-time transparency, and scalable governance, positioning them for sustainable institutional adoption. Table 1 lists the seven critical success factors.

Table 1: Summary Table of Critical Success Factors

Critical Success Factor	Core Content
1. Supra-sovereign collateral quality	Gold provides low volatility, no default risk, global liquidity, and physical verifiability
2. Regulatory certainty under Hong Kong's *Stablecoin Ordinance*	Licensing, 100% reserve, segregation, audit, and legal clarity for commodity-backed stablecoins
3. Trust triangle: full collateral + segregation + audit	100%+ physical gold backing, third-party custody, dual on/off-chain verification

Critical Success Factor	Core Content
4. Price stability: oracle + arbitrage + redemption	Decentralized pricing, dynamic over-collateral, arbitrage loop, circuit breakers
5. Hong Kong's locational-ecological synergy	Gold hub + offshore finance + digital asset center + cross-border connectivity
6. Full-cycle multi-layer risk governance	Anti-run, anti-volatility, compliance, cybersecurity, and operational safeguards
7. Compliance-technology integrated governance	Regulatory rules coded in smart contracts; automated compliance and audit

6. Contributions and Implications

6.1 Contributions to Research

This paper makes three significant academic contributions: first, it develops and validates a seven-dimensional critical success factor framework specifically for gold-backed stablecoins, addressing a major gap in research focused on fiat or algorithmic models with a theoretically grounded, empirically informed, and structurally replicable framework; second, it reveals the unique synergistic institutional logic that allows an offshore financial center like Hong Kong to combine gold market infrastructure, financial openness, and progressive stablecoin regulation to foster compliant innovation, extending theories of offshore financial competitiveness into the digital asset era; third, it establishes an integrated paradigm of asset–regulation–mechanism–ecosystem–risk–technology that unifies fragmented literature on collateral quality, governance, price stability, and cross-border finance, providing a structured foundation for future research on commodity-pegged stablecoins worldwide.

6.2 Practical Implications

Practical implications apply to issuers, regulators, gold providers, investors, financial institutions, and Hong Kong's broader economy. For stablecoin issuers, success requires strict adherence to full collateralization, transparent reserve management, reliable redemption mechanisms, and compliance-by-design principles, with prioritization of independent auditing, real-time disclosure, and robust custody to build and maintain market trust [21][39]. For regulators, particularly the Hong Kong Monetary Authority, maintaining regulatory stability, enhancing cross-border coordination, and supporting responsible gold tokenization and real-world asset development will strengthen Hong Kong's position as a leading digital asset hub, while promoting interoperability and clarity to encourage institutional participation[15][8]. For Hong Kong as a global financial center, Gold-Backed Stablecoins offer a strategic opportunity to reinforce its dual status as a premier digital asset center and a global gold pricing and trading hub, attracting global issuers, investors, and partners to drive innovation, employment, and capital inflows by leveraging triple ecosystem advantages [20][1].

For gold suppliers, vault operators, and financial institutions, Gold-Backed Stablecoins create new demand for secure storage, auditing, institutional-grade custody, and structured investment products, expanding revenue streams and strengthening the entire gold-finance ecosystem. For retail and institutional investors, Gold-Backed Stablecoins offer a stable, secure, and portable store of value, combining gold's safe-haven benefits with blockchain efficiency and accessibility. Table 2 outlines the practical implications for the stakeholders.

Table 2. Summary table for the stakeholders and their respective roles

Stakeholder	Role & Expected Actions / Value from GBSCs
Stablecoin Issuers	Maintain ≥100% gold collateral; independent audit; real-time redemption; comply with HK's *Stablecoin Ordinance*

Stakeholder	Role & Expected Actions / Value from GBSCs
Regulators (HKMA)	Maintain stable rules; improve cross-border regulatory coordination; support gold tokenization and RWA
Gold Suppliers / Vaults	Provide secure physical storage; support verification; enhance transparency
Institutional Investors	Access safe-haven digital assets; improve portfolio diversification; low-volatility tool
Retail Users	Stable value storage; anti-inflation; cross-border transfer without high volatility
Financial Institutions	New structured products; liquidity tools; gold-backed digital investment services
Hong Kong Economy	Strengthen gold hub status; enhance digital financial competitiveness; attract global capital

7. Conclusion, Limitation and Future Research

This paper identifies seven critical success factors for gold-backed stablecoins in Hong Kong, including asset quality, regulation, and ecosystem synergy, filling research gaps and offering implications for policymakers and issuers.

This study faces three main limitations that point to meaningful directions for future research. First, since gold-backed stablecoins remain in an early stage of development in Hong Kong, longitudinal and large-sample empirical data are limited, restricting robust quantitative testing of causal relationships as most evidence remains theoretical or qualitative and reduces statistical generalizability. Second, the study focuses on Hong Kong's institutional context, and cross-jurisdictional comparisons with other gold hubs or regulatory regimes, such as Singapore, the UAE, or Switzerland, are not fully explored, thereby limiting insights into international competitiveness. Third, the rapid evolution of technology, regulation, and market structure means that some success factors may shift over time, especially as AI-driven oracle systems, institutional decentralized finance, and cross-border stablecoin agreements mature.

Future research can extend this study to validate the critical success factors and determine their relative weight. A follow-up study will employ an anonymous Delphi questionnaire survey. The expert panel will cover domains including the gold market, digital asset regulation, blockchain technology, offshore finance, and institutional custody. A 5-point Likert scale will be used for scoring, with a coefficient of variation less than 0.25 adopted as the criterion for consensus convergence. Through the Delphi method, factors will be revised, scored, and ranked by weight. The interactions and systematic logic among factors will be further analysed, and the framework's boundaries of applicability and generalisability will eventually be evaluated. By conducting large-sample empirical analysis once more, Gold-Backed Stablecoin projects launch in Hong Kong using panel data to test factor significance and magnitude, undertake comparative institutional analysis between Hong Kong and other digital asset centers to identify relative advantages and policy optimizations, and explore innovative applications in cross-border trade finance, offshore Renminbi settlements, and central bank reserve diversification to open new practical impact avenues. As markets mature, longitudinal studies will further clarify how success factors evolve and interact over time.

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